STA2002 Tutorial 9

1. Let $X_1, X_2, \cdots, X_n$ be a random sample from $N\left(0, \sigma^2\right)$.

   (a) Show that $C = \left\{(x_1, x_2, \cdots, x_n) : \sum_{i=1}^n x_i^2 \geq c\right\}$ is a best critical region for testing $H_0 : \sigma^2 = 4$ against $H_1 : \sigma^2 = 16$

   (b) If $n = 15$, find the value of c so that $\alpha = 0.05$.

   (c) If $n = 15$ and c is the value found in part (b), find the approximate value of $\beta = P\left(\sum_{i=1}^n = X_i^2 < c; \sigma^2 = 16\right)$.

   (a)

$$\frac{L(4)}{L(16)} = \frac{(1/2\sqrt{2\pi})^n \exp\left[-\Sigma x_i^2/8\right]}{(1/4\sqrt{2\pi})^n \exp\left[-\Sigma x_i^2/32\right]} = 2^n \exp\left[-3\Sigma x_i^2/32\right] \leq k$$

$$-\frac{3}{32}\sum_{i=1}^n x_i^2 \leq \ln k - \ln 2^n$$

$$\sum_{i=1}^n x_i^2 \geq -\left(\frac{32}{3}\right)(\ln k - \ln 2^n) = c$$

   (b)

$$0.05 = P\left(\sum_{i=1}^{15} X_i^2 \geq c; \sigma^2 = 4\right) = P\left(\frac{\sum_{i=1}^{15} X_i^2}{4} \geq \frac{c}{4}; \sigma^2 = 4\right)$$

   Thus

$$\frac{c}{4} = \chi^2_{0.05}(15) = 25 \text{ and } c = 100$$

   (c)

$$\beta = P\left(\sum_{i=1}^{15} X_i^2 < 100; \sigma^2 = 16\right)$$

$$= P\left(\frac{\sum_{i=1}^{15} X_i^2}{16} < \frac{100}{16} = 6.25\right) \approx 0.025$$

2. Assume that the weight X in ounces of a "10-ounce" box of cornakes is $N(\mu, 0.03)$. Let $X_1, X_2, \cdots, X_n$ be a random sample from this distribution.

   (a) To test the hypothesis $H_0 : \mu \geq 10.35$ against the alternative hypothesis $\mathrm{H}_1 : \mu < 10.35$, what is the critical region of size $\alpha = 0.05$ specied by the likelihood ratio test criterion?

   (b) If a random sample of $n = 50$ boxes yielded a sample mean of $\bar{x} = 10.31$, is H_0 rejected?

   (c) What is the p-value of this test?

   (a) If $\bar{x} \geq 10.35, \lambda = 1$; but if $\bar{x} < 10.35$, then
   $\frac{\bar{x}-10.35}{\sqrt{0.03/n}} \leq -\sqrt{-2\ln k} = -z_{0.05} = -1.645$

$$\lambda = \frac{L(\hat{\omega})}{L(\Omega)} = \frac{[1/(\sqrt{0.06\pi})]^{n/2} \exp\left[-\sum_{i=1}^n (x_i - 10.35)^2/(0.06)\right]}{[1/(\sqrt{0.06\pi})]^{n/2} \exp\left[-\sum_{i=1}^n (x_i - \bar{x})^2/(0.06)\right]} \leq k$$

$$\exp\left[-\frac{n}{0.06}(\bar{x} - 10.35)^2\right] \leq k$$

$$-\frac{n}{0.06}(\bar{x} - 10.35)^2 \leq \ln k$$

(b)

$$\frac{10.31-10.35}{\sqrt{0.03/50}}=-1.633>-1.645$$

do not reject H_0

(c)

$$P(Z\leq -1.633)=0.0513$$

3. Let $X_1, X_2, \cdots, X_n$ be a random sample from an exponential distribution with mean θ. Show that the likelihood ratio test of $H_0:\theta=\theta_0$ against $H_1:\theta\neq\theta_0$ has a critical region of the form $\sum_{i=1}^n x_i\leq c_1$ or $\sum_{i=1}^n x_i\geq c_2$ How would you modify this test so that chi-square tables can be used easily?

In $\Omega, \widehat{\mu}=\bar{x}$. Thus,

$$\lambda=\frac{(1/\theta_0)^n\exp\left[-\sum_1^n x_i/\theta_0\right]}{(1/\bar{x})^n\exp\left[-\sum_1^n x_i/\bar{x}\right]}\leq k$$

$$\left(\frac{\bar{x}}{\theta_0}\right)^n\exp\left[-n\left(\bar{x}/\theta_0-1\right)\right]\leq k$$

Plotting λ as a function of $w=\bar{x}/\theta_0$, we see that $\lambda=0$ when $\bar{x}/\theta_0=0$, it has a maximum when $\bar{x}/\theta_0=1$, and it approaches 0 as $\bar{x}/\theta_0$ becomes large. Thus $\lambda\leq k$ when $\bar{x}\leq c_1$ or $\bar{x}\geq c_2$

Since the distribution of $\frac{2}{\theta_0}\sum_{i=1}^n X_i$ is $\chi^2(2n)$ when H_0 is true, we could let the critical region be such that we reject H_0 if

$$\frac{2}{\theta_0}\sum_{i=1}^n X_i\leq\chi^2_{1-\alpha/2}(2n)\quad\text{or}\quad\frac{2}{\theta_0}\sum_{i=1}^n X_i\geq\chi^2_{\alpha/2}(2n)$$

4. Let X and Y equal, respectively, the blood volumes in milliliters for a male who is a paraplegic and participates in vigorous physical activities and for a male who is able-bodied and participates in everyday, ordinary activities. Assume that $X\sim N\left(\mu_X,\sigma_X^2\right)$ and $Y\sim N\left(\mu_Y,\sigma_Y^2\right)$. Following are $n=7$ observations of X :

1612	1352	1456	1222	1560	1456	1924

Following are $m=10$ observations of Y :

1082	1300	1092	1040	910
1248	1092	1040	1092	1288

$$s_p^2=29046.52, s_x^2=49669.905, s_y^2=15297.6, t_{0.025}(15)=2.1314$$

Use the observations of X and Y to

(a) Give a point estimator for $\mu_X-\mu_Y$

(b) Find the 95% confidence interval for $\mu_X-\mu_Y$.

(a) Give a point estimator for $\mu_X-\mu_Y$

$$\bar{x}-\bar{y}=1511.714-1118.400=393.314$$

(b) First of all, we need to test whether these two unknown variance are equal or not, only variance are equal , we could use the pooled T.
We are interesting to test :

$$H_0:\sigma_X^2=\sigma_Y^2,\quad H_1:\sigma_X^2\neq\sigma_Y^2.$$

Under H_0, we have the test statistic

$$\frac{S_X^2}{S_Y^2}\sim F_{(n-1,m-1)}$$

we reject if the test statistic F is too large:

$$F\geq F_{\alpha/2}(n-1,m-1)$$

or if the test statistic F is too small

$$F \leq F_{1-\alpha/2}(n-1, m-1) = \frac{1}{F_{\alpha/2}(m-1, n-1)}$$

Here we have $s_x^2 = 49,669.905, s_y^2 = 15,297.600$. And we check the F table to get the critical value is

$$F_{0.025}(7-1, 10-1) = 4.32, F_{0.975}(6, 9) = \frac{1}{F_{0.025}(9, 6)} = 0.1812$$

And the ratio of sample variance is 3.2469 which is not in the critical region. So we do not reject the null hypothesis.

Since the variances σ_X^2 and σ_Y^2 are equal, we use pooled T. We have

$$\frac{\bar{X} - \bar{Y} - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t(m+n-2).$$

We have $S_p^2 = 29046.52$. Thus the 95% C.I. for $\mu_X - \mu_Y$ is

$$[214.2956, 572.3324].$$